

AI HR STRATEGY ENGINE – BACKEND ARCHITECTURE MAP

Note

This is a high-level backend system map.

AI HR Strategy Engine is a **governing intelligence layer**, not a reporting tool.

Final scope, timelines, and cost depend on phase prioritization.

1. CORE SERVICES

A. API Gateway

- Request routing
 - Authentication & Authorization
 - Rate limiting
 - Request logging
 - CORS handling
 - Payload validation
 - Tenant & strategy isolation
 - Strategy execution audit middleware
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B. Authentication Service

- User login / logout
 - JWT token generation
 - Role-based access control (HR, Executive, Manager, Strategy Owner, Auditor)
 - Session management
 - Strategy approval permissions
 - Read vs Execute access enforcement
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2. BUSINESS LOGIC SERVICES

A. Strategy Definition Service (Strategy-as-Data)

Endpoints

POST /api/strategy

GET /api/strategy

GET /api/strategy/{id}

PUT /api/strategy/{id}

POST /api/strategy/{id}/version

Business Logic

- Store HR strategy as **structured objects**, not documents
 - Strategy composed of:
 - Strategic pillars
 - Strategic initiatives
 - KPIs
 - Roadmap phases
 - Full strategy versioning
 - Editable without regeneration
 - Immutable historical versions
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B. Strategic Pillars Service

Endpoints

POST /api/strategy/pillars

PUT /api/strategy/pillars/{id}

DELETE /api/strategy/pillars/{id}

GET /api/strategy/pillars

Business Logic

- Pillars defined dynamically (not hard-coded)
 - Pillar attributes:
 - Name
 - Description
 - Priority
 - Pillars reorderable
 - Pillars linked to business objectives
 - Reusable across strategy versions
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C. Strategic Initiatives Service (Executable Strategy Layer)

Endpoints

POST /api/strategy/initiatives

PUT /api/strategy/initiatives/{id}

GET /api/strategy/initiatives

PATCH /api/strategy/initiatives/{id}/status

Business Logic

- Each initiative linked to:
 - One strategic pillar
 - One or more KPIs
 - Initiative attributes:
 - Objective
 - Scope
 - Owner
 - Timeline
 - Status tracking (Planned / Active / Completed)
 - Initiative dependency management
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D. Strategy–Execution Linkage Service

Endpoints

POST /api/strategy/initiatives/{id}/link-module

GET /api/strategy/initiatives/{id}/execution-status

Business Logic

- Link initiatives to execution engines:
 - Recruitment & Mobility
 - Performance Management
 - TNA
 - Workforce Planning
 - Configurable linkage (per initiative)
 - Execution feedback loop into strategy
 - Broken / inactive links flagged

E. Strategy Readiness Scoring Service

Endpoints

POST /api/strategy/readiness/calculate

GET /api/strategy/initiatives/{id}/readiness

Business Logic

- Readiness score per initiative (0–100)
- Factors:
 - Organizational maturity (OMA)
 - Workforce capability
 - Cultural readiness
 - Resource availability
 - Scoring logic configurable
 - Stored with initiative version

F. Strategy Confidence Engine

Endpoints

GET /api/strategy/confidence

Business Logic

- Aggregate readiness scores
- Assign confidence level:
 - High
 - Medium
 - Low
 - Recalculated on any data change
 - Executive-facing confidence signal

G. Strategy Benchmarking Service

Endpoints

POST /api/strategy/benchmark

GET /api/strategy/benchmark/report

Business Logic

- Benchmark strategy against:
 - Industry standards
 - Company size
 - Region
 - Pillar-level comparison
 - Variance detection (Above / At / Below)
 - Benchmark dataset versioning
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H. Strategy Explainability & Risk Service

Endpoints

GET /api/strategy/explainability

GET /api/strategy/initiatives/{id}/risk

Business Logic

- Explain:
 - Why pillars were selected
 - What data influenced initiatives
 - Business-language explanations
 - AI reasoning traceability
 - Risk of non-execution analysis:
 - Operational
 - Talent
 - Financial
 - Severity scoring
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3. INTEGRATION SERVICES

A. Organizational Maturity Integration (OMA)

- Fetch maturity scores
 - Domain-level gaps
 - Feed readiness scoring
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B. Workforce Analytics Integration

- Headcount
 - Skills inventory
 - Attrition trends
 - Capability heatmaps
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C. Execution Engines Integration

- Recruitment Engine
 - Performance Management
 - TNA
 - Internal Mobility
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D. AI Proxy Service

- Strategy synthesis
 - Gap analysis
 - Readiness scoring
 - Explainability generation
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E. Notification Service

- Strategy approval alerts
 - Confidence change notifications
 - Execution risk alerts
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4. DATABASE (SUMMARY)

- HR_Strategies
- Strategy_Versions
- Strategic_Pillars

- Strategic_Initiatives
 - Initiative_Dependencies
 - Strategy_KPIs
 - Strategy_Roadmaps
 - Strategy_Readiness_Scores
 - Strategy_Confidence_Levels
 - Strategy_Benchmarks
 - Strategy_Execution_Links
 - Strategy_Explainability
 - Strategy_Risk_Analysis
 - Audit_Logs
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DATABASE ARCHITECTURE

A. HR_Strategies

- id (PK)
 - organization_id
 - current_version_id
 - status (Draft / Active / Archived)
 - created_by
 - created_at
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B. Strategy_Versions

- id (PK)
 - strategy_id (FK)
 - version_number
 - created_at
-

C. Strategic_Pillars

- id (PK)
- strategy_version_id (FK)
- name
- description
- priority

D. Strategic_InitiatIVES

- id (PK)
- pillar_id (FK)
- objective
- owner
- timeline_start
- timeline_end
- status

E. Strategy_Readiness

- id (PK)
- initiative_id (FK)
- readiness_score
- scoring_breakdown (JSON)

F. Strategy_Confidence

- id (PK)
- strategy_version_id (FK)
- confidence_level
- calculated_at

G. Strategy_Benchmarks

- id (PK)
- strategy_version_id (FK)
- benchmark_type (Industry / Size / Region)
- variance_data (JSON)

H. Strategy_Execution_Links

- id (PK)
- initiative_id (FK)

- module_name
 - execution_status
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I. Strategy_Explainability

- id (PK)
 - entity_type (Pillar / Initiative)
 - entity_id
 - explanation_text
 - data_sources (JSON)
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J. Strategy_Risks

- id (PK)
 - initiative_id (FK)
 - risk_type
 - severity
 - impact_description
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USER STORY COVERAGE

- ☒ EPIC 1 – Strategy-as-a-System
- ☒ EPIC 2 – Readiness & Confidence Scoring
- ☒ EPIC 3 – Strategy Benchmarking
- ☒ EPIC 4 – Explainability & Risk Transparency